

Suhyeon Lee

Seoul, Republic of Korea * orion-alpha_at_korea.ac.kr * <https://suhyeonlee.com>

PROFESSIONAL SUMMARY

Blockchain Security Researcher specializing in **Protocol Design** and **Game-Theoretic Analysis** within distributed systems. Currently working as a Security Researcher at **Tokamak Network** and as a Visiting Research Fellow at **Hashed Open Research**, a leading crypto-VC backed institute. My research focuses on the intersection of Layer 2 scalability, incentive compatibility, and strategic security analysis.

With a background in **cyber defense engineering** from the **Agency for Defense Development (ADD)**—South Korea's primary defense research institution—and the ROK Cyber Operations Command, I bring experience in **designing secure operating systems, cyber threat intelligence, and attack and defense strategies**. I am passionate about applying these rigorous theoretical models and experiences to solve practical challenges in blockchain consensus and cryptoeconomics.

EDUCATION

Sep. 2017 – Feb. 2023 **Ph.D. in Cybersecurity**
Department of Information Security, School of Cybersecurity, **Korea University**
Supervisor: Prof. Seungjoo Kim ()
Thesis: **Revisiting Conventional Wisdom in PoW and PoS-based Blockchains**

Mar. 2012 – Feb. 2016 **B.Eng. in Cyber Defense**
Department of Cyber Defense, **Korea University**
Supervisor: Prof. Jong In Lim (jilim@korea.ac.kr)

PROFESSIONAL EXPERIENCE

Nov. 2025 – **Visiting Research Fellow**
 Hashed Open Research
- Contributing to project-based research and reports on blockchain and cryptocurrency

June. 2023 – **Security Researcher**
 Tokamak Network
- Security research focusing on Layer 2 blockchain systems

Sep. 2024 – Aug. 2025 **Adjunct Professor**

School of Cyber Security, Korea University

- Teaching a regular lecture SCS606 “Blockchain and Cryptocurrency”

Jan. 2023 – May. 2023 **AI Technology Research Officer / Army Captain**

ROK Cyber Operations Command

- Led cyber defense research using AI technologies

Jan. 2021 – Dec. 2022 **Defense System Development Officer / Army Captain**

ROK Cyber Operations Command

- Led cyber defense system design and development.
- Completed cyber active defense training focusing on infiltration.

Jul. 2019 – Dec. 2020 **Cryptography Technology Research Officer / Army Captain**

ROK Cyber Operations Command

- Participated in Korea-US joint military exercises.
- Worked for cooperation with relevant security agencies.
- Planned multiple cyber defense projects to support cyber operations.

Jul. 2016 – Jul. 2019 **Researcher / Army Lieutenant**

2nd R&D Institute, Agency for Defense Development (ADD)

- Participated in R&D for collecting and processing Cyber Threat Intelligence (CTI).
- Participated in R&D for development of secure Real Time Operation System (RTOS) for weapon systems with high security verification ($EAL \geq 4$).

WORKING PAPERS (Preprint)

S. Lee, and Y. Park, “**Looking for Attention: Attention Test Design for Validator Monitoring in Optimistic Rollups**,” arXiv:2505.24393, June 2025.

S. Lee, and H. Kim, “**Inside Qubic’s Selfish Mining Attack in Monero: Evidence, Tactics, and Limits.**”

PUBLICATIONS - INTERNATIONAL CONFERENCES

S. Lee, D. Nguyen, and D. Lee, “(Im)possibility of Incentive Design for Challenge-based Blockchain Protocols,” In Proceedings of the *10th Workshop on Trusted Smart Contracts (WTSC 2026)*, in association with *Financial Cryptography 2026*, pp. TBD, St. Kitts, Saint Kitts and Nevis, March 6, 2026.

S. Lee, “180 Days After EIP-4844: Will Blob Sharing Solve Dilemma for Small Rollups?”, In Proceedings of the *DICG Workshop*, in association with the *45th IEEE International Conference on Distributed Computing Systems (ICDCS 2025)*, pp. TBD, Glasgow, UK, July 20–23, 2025.

S. Lee, and E. Gee, “Commit-Reveal²: Randomized Reveal Order Mitigates Last-Revealer Attacks in Commit-Reveal,” In Proceedings of the *IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2025)*, pp. TBD, Pisa, Italy, June 2-6, 2025.

S. Lee, “Hollow Victory: Analysis on Challenger's Incentive in Optimistic Rollup Dispute Game,” In Proceedings of the *9th Workshop on Trusted Smart Contracts (WTSC 2025)*, in association with *Financial Cryptography 2025*, pp. TBD, Miyakojima, Japan, April 18, 2025.

H. Yang, **S. Lee**, and S. Kim, “A Tip for IOTA Privacy: IOTA Light Node Deanonymization via Tip Selection,” In Proceedings of the International Conference on *IEEE Blockchain Conference (ICBC 2024)*, Dublin, Ireland, May 2024.

S. Lee, and S. Kim, “Do You Really Need to Disguise Normal Servers as Honeypots?,” In Proceedings of the 40th International Conference on *IEEE Military Communications Conference (MILCOM 2022)*, pp. 166-172, Rockville, MD, US, November 2022.

S. Lee, D. Lee, and S. Kim, “Block Double-Submission Attack: Block Withholding Can Be Self-Destructive,” In Proceedings of the 4th International Conference on *ACM Advances in Financial Technologies (AFT '22)*, Cambridge, MA, US, September 2022.

S. Lee and S. Kim, “Proof-of-Stake at Stake: Predatory, Destructive Attack on PoS Cryptocurrencies,” In Proceedings of *ACM MobiCom 2020 Workshops* – the 3rd International Workshop on Cryptocurrencies and Blockchains for Distributed Systems (*CryBlock '20*), pp. 7-11, Virtual Conference, September 2020.

S. Lee and S. Kim, “Blockchain for Cyber Defense: Will It Be as Good as You Think?” the 28th *DEF CON Hacking Conference (DEFCON 28) – the 2nd Blockchain Village*, Virtual Conference, August 2020.

S. Lee and S. Kim, “Countering Block Withholding Attack Efficiently,” In Proceedings of *IEEE INFOCOM 2019 Workshops* – the 2nd International Workshop on Cryptocurrencies and Blockchains for Distributed Systems (*CryBlock '19*), pp. 330-335, Paris, France, May 2019.

PUBLICATIONS – INTERNATIONAL JOURNALS

S. Lee, E. Jee, J. Lee, “**Implementation of Cost-Effective Verification for Pietrzak's Verifiable Delay Function in Ethereum Smart Contracts**,” *Blockchain: Research and Applications* (I/F 5.6), In Press, June 2025.

S. Lee, **S. Lee**, J. Park, K. Kim, and Lee, K, “**Hiding in the Crowd: Ransomware Protection by Adopting Camouflage and Hiding Strategy With the Link File**,” *IEEE Access* (I/F 3.476), vol. 11, pp. 92693- 92704, 2023.

S. Lee and S. Kim, “**Shorting Attack: Predatory, Destructive Short Selling on Proof-of-Stake Cryptocurrencies**,” *Concurrency and Computation: Practice and Experience* (I/F 2), vol. 35, Issue 16, e6585, July 2023.

S. Lee and S. Kim, “**Rethinking Selfish Mining under Pooled Mining**,” *ICT Express* (I/F 4.754), vol. 9, Issue 3, pp. 356- 361, June 2023.

S. Lee and S. Kim, “**Blockchain as a Cyber Defense: Opportunities, Applications, and Challenges**,” *IEEE Access* (I/F 3.476), vol. 10, pp. 2602-2618, December 2021.

S. Lee, H. K. Kim, and K. Kim, “**Ransomware Protection using the Moving Target Defense Perspective**,” *Computers & Electrical Engineering* (I/F 4.152), vol. 78. pp. 288-299, September 2019.

PATENTS

S. Kim and **S. Lee**, “**Method for generating to tag for tracking document**,” submitted, May 2022.

H. Kim, K. Kim, and **S. Lee**, “**Ransomware damage prevention method through extension randomization, recording medium and apparatus for performing the method**,” no. 10-2316404, October 2021.

S. Kim and **S. Lee**, “**A block withholding attack detection device and its operation method**,” no. 10-2122824, June 2020.

S. Kim and **S. Lee**, “**Mining pool management server capable of suppressing selfish mining strategies of competitive mining pools and method of operation**,” no. 10-2154354, September 2019.

D. Lee, B. Ko, C. Lim, D. Kim, H. Ryu, **S. Lee**, and J. Park, “**Device and method for extracting web page information through inference and learning of html tag information**,” no. 10-2018-0137256, December 2018.

INVITED TALKS

Suhyeon Lee, “**Research and Experience in Cyber Defense**,” Capstone Project in the Department of Cyber Defense, Korea University, Seoul, April 2023.

Suhyeon Lee, “**Threat Modeling for Cybersecurity of Industrial Products**,” 78ResearchLab, Gangnam-gu, Seoul, October 2022.

Suhyeon Lee, “Shorting Attack on Proof-of-Stake Cryptocurrencies,” Monthly Crypto Seminar in Ripple, Virtual, July 2022.

Suhyeon Lee, “Blockchain Technology for Military Cyber Security,” The 28th Network Security Conference Korea (NetSec-KR), COEX, Seoul, Korea, April 2022.

Suhyeon Lee, “K-Cyber Security Challenge Winner’s Talk,” Conference Center of the Federation of Korean Industries, Seoul, December 2019.

LANGUAGE & ADDITIONAL SKILLS

- **Language:** Korean (native), English (proficient)
- **Programming:** C, Python (familiar)
- **Certification:** Certified Information Systems Security Professional (**CISSP**), Certification for English Mediated Instruction (Awarded by the Center for Teaching and Learning of Korea University)

AWARDS & HONORS

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| Dec. 2019 | 1st Prize in Korea Security Data Challenge (AI Section)
Granted by Korea Internet & Security Agency (KISA) |
| 2012 – 2016 | Full-tuition Military Scholarship
Granted by Ministry of National Defense, Republic of Korea |
| 2012 – 2016 | Merit-based Scholarship
Granted by Korea University, Republic of Korea |
| 2017 – 2019 | General Scholarships
Granted by Korea University, Republic of Korea |

REFERENCES

Professor. Seungjoo Kim (PhD Supervisor)

- School of Cybersecurity, Korea University, Republic of Korea
- President of Office of Digital Information, Korea University
- Member of Republic of Korea Presidential Defense Innovation Committee
- Email: skim71_at_korea.ac.kr

Professor. Huy Kang Kim

- School of Cybersecurity, Korea University, Republic of Korea

Edited on 2026-02-28

- Director of Hacking and Countermeasure Research Lab (<https://ocslab.hksecurity.net>)
- Email: cenda_at_korea.ac.kr

Senior Researcher. Donghwan Lee

- Agency for Defense Development, Republic of Korea
- Email: dlee_at_add.re.kr

Chair-Professor. Jongin Lim

- School of Cybersecurity, Korea University, Republic of Korea
- Advisor of Kim & Chang Law Firm, Republic of Korea
- Former Special Advisor to the President for National Security
- Email: jilim_at_korea.ac.kr